Cross-Boundary Trace-Class Estimates for Wilson–Dirac Quasifree States

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Abstract. We derive lattice Fourier moment identities, two-boundary Laplace and modified-Bessel representations, trace-class convergence and subcritical fractional Sobolev nuclear bounds for compressed infinite-chain Wilson–Dirac projectors. The finite quasifree trace-path identity and these estimates prove an entropy-difference convergence rate for this specified model. These results alone do not establish the complete Wilson relative-entropy asymptotic or its modular remainder.

Model and definitions

The model is the infinite-chain Wilson–Dirac occupied-band projection

$$P_m(k) = \frac{1}{2}\left[I - \frac{\sin k\,\sigma_x + (1+m-\cos k)\sigma_z}{\sqrt{m^2 + 4(1+m)\sin^2(k/2)}}\right], \qquad m = \mu/L.$$

The finite region consists of **L cells, each with two orbitals**. Restricting the *infinite-chain projection* to these cells is essential; the proof is not for the ground-state projection of an open finite chain. Let $U_L\colon \ell^2(\mathbb{Z};\mathbb{C}^2) \to L^2(\mathbb{R};\mathbb{C}^2)$ be the isometry whose outputs are constant on each interval $[j/L, (j+1)/L)$, and $\chi = 1_{(0,1)}$. Set

$$\begin{aligned}
\widetilde{P}_{L,\mu} &= U_L \mathsf{P}_{\mu/L} U_L^*, & X_{L,\mu} &= \chi \widetilde{P}_{L,\mu}(1-\chi),\\
P^c_\mu &= \mathcal{F}^{-1}\frac{1}{2}\left[I - \frac{u\sigma_x + \mu\sigma_z}{\sqrt{u^2+\mu^2}}\right]\mathcal{F}, & X_\mu &= \chi P^c_\mu(1-\chi),\\
Z_{L,\mu} &= X_{L,\mu} - X_{L,0}, & Z_\mu &= X_\mu - X_0,\\
\mathbb{D}_\mu &= \begin{pmatrix} 0 & Z_\mu \\ Z_\mu^* & 0 \end{pmatrix}.
\end{aligned}$$

The value of the continuum multiplier at $(u,\mu) = (0,0)$ is irrelevant. The L^2 decomposition is into the interior and **both** exterior half-lines.

Lattice Fourier moment identity (no contour deformation)

In the fixed σ_y eigenbasis, the Bloch projection has off-diagonal function

$$q_\tau(e^{ik}) = \frac{1+m-e^{ik}}{E_m(k)} = \left(1-\tau e^{ik}\right)^{1/2}\left(1-\tau e^{-ik}\right)^{-1/2}, \qquad \tau = (1+m)^{-1}.$$

For $0 < \tau < 1$ both binomial series converge absolutely and uniformly on the unit circle. Put $c_j = (-1/2)_j/j!$ and $b_j = (1/2)_j/j!$. For integer $d \geq 1$, the Fourier convention $q_d = (2\pi)^{-1}\int_{-\pi}^{\pi} e^{idk} q_\tau(e^{ik})\,dk$ yields

$$q_d(\tau) = \tau^d \sum_{j\geq 0} c_j b_{j+d} \tau^{2j} = \frac{\tau^d}{\pi} B(d+\tfrac{1}{2}, \tfrac{1}{2}) {}_2F_1(-\tfrac{1}{2}, d+\tfrac{1}{2}; d+1; \tau^2),$$

$$q_{-d}(\tau) = \tau^d \sum_{j\geq 0} c_{j+d} b_j \tau^{2j} = -\frac{\tau^d}{\pi} B(d-\tfrac{1}{2}, \tfrac{3}{2}) {}_2F_1(\tfrac{1}{2}, d-\tfrac{1}{2}; d+1; \tau^2).$$

The prefactors follow from $b_d = B(d+1/2, 1/2)/\pi$ and $c_d = -B(d-1/2, 3/2)/\pi$. Euler's beta integral, or termwise integration of its binomial expansion, therefore proves exactly

$$\boxed{\begin{aligned} q_d(\tau) &= \frac{1}{\pi} \int_0^\tau x^{d-1} \sqrt{\frac{x(1-\tau x)}{\tau - x}}\, dx, \\ q_{-d}(\tau) &= -\frac{1}{\pi} \int_0^\tau x^{d-1} \sqrt{\frac{\tau - x}{x(1-\tau x)}}\, dx. \end{aligned}} \tag{B1}$$

The integrands have integrable endpoints for $d \geq 1$; dominated convergence extends the result to $\tau = 1$, where the densities become $\sqrt{x}$ and $x^{-1/2}$. This is a separate derivation of the sign-sensitive positive- and negative-displacement formula, not an assumption about a contour.

Both boundary channels and direct continuum identification

For $\mu > 0$ set

$$a = a_L = L \log(1 + \mu/L), \quad \tau = e^{-a/L},$$

$$R_L(t) = \sqrt{\frac{\sinh((t+a)/(2L))}{\sinh((t-a)/(2L))}}, \quad R_a(t) = \sqrt{\frac{t+a}{t-a}}, \quad t > a.$$

Define $d_L^{\pm}(t) = 1_{t>a} R_L(t)^{\pm 1} - 1$ and $d_a^{\pm}(t) = 1_{t>a} R_a(t)^{\pm 1} - 1$. For $x \in (0,1),\ y > 0$, let

$$u_{L,t}(x) = e^{-t\lfloor Lx \rfloor / L}, \quad v_{L,t}(y) = e^{-t\lfloor Ly \rfloor / L}, \quad u_t(x) = e^{-tx}, \quad v_t(y) = e^{-ty}.$$

Writing $P_L(t) = |u_{L,t}\rangle\langle v_{L,t}|$ and $P(t) = |u_t\rangle\langle v_t|$, substitution $x = e^{-t/L}$ in (B1) gives **exactly**, with no asymptotic replacement,

$$H_L^+ = \int_0^\infty e^{-3t/(2L)} d_L^+(t) P_L(t)\, dt, \qquad H_L^- = \int_0^\infty e^{-t/(2L)} d_L^-(t) P_L(t)\, dt. \tag{B2}$$

The unequal exponents $3/2$ and $1/2$ result from the alternative square-root channel and the Jacobian $dx = -e^{-t/L} dt/L$. They cannot be interchanged. In the σ_y basis, the left exterior contribution to $Z_{L,\mu}$ is $(2\pi)^{-1} \left(\begin{smallmatrix} 0 & -H_L^+ \\ H_L^- & 0 \end{smallmatrix}\right)$. Reflecting the interior interval gives the right exterior contribution, with channel signs exchanged and the same $(2\pi)^{-1}$ factor. The triangle inequality over both exterior columns gives the factor $1/\pi$ in the assembled trace-norm estimate, *not* $1/(2\pi)$.

The corresponding continuum candidates are $H_b^{\pm} = \int_0^\infty d_b^{\pm}(t) P(t) dt$, with $d_b^{\pm} = 1_{t>b}[(t+b)/(t-b)]^{\pm 1/2} - 1$. These are trace-class Bochner integrals, by the estimates established below.

An additional kernel check follows. For a separation $r > 0$, substitute $t = b \cosh v$. Using $\coth(v/2) \sinh v = \cosh v + 1$ and $\tanh(v/2) \sinh v = \cosh v - 1$, the classical integral $K_j(z) = \int_0^\infty e^{-z \cosh v} \cosh(jv) dv$ yields

$$\boxed{\begin{aligned}\int_0^\infty d_b^+(t)e^{-rt}dt &= b[K_1(br)+K_0(br)]-\frac{1}{r},\\ \int_0^\infty d_b^-(t)e^{-rt}dt &= b[K_1(br)-K_0(br)]-\frac{1}{r}.\end{aligned}} \tag{B3}$$

The Fourier kernel of $q_b(u) = (b-iu)/\sqrt{u^2+b^2}$ is $b[K_0(br)+K_1(br)]/\pi$ for $r>0$, while the massless Fourier kernel is $1/(\pi r)$. For negative displacement, the K_1 term changes sign and matches the sign-reversed second channel. Equation (B3) thus identifies the continuum Laplace candidate with the **specified continuum Dirac projector** rather than merely defining a formal limiting kernel. This uses standard Bessel integral identities, without invoking a lattice branch-contour argument.

Finally, the full-line step projections converge strongly on band-limited test functions to P_b^c because the rescaled Bloch multipliers converge pointwise almost everywhere, the cell form factor tends to one, and projectors are uniformly bounded. Density and the identity $\|P_n f - Pf\|^2 = \langle f, (P_n + P - P_nP - PP_n)f\rangle$, or the standard weak-to-strong argument for projections, extend this convergence to all L^2. The trace-norm limit of the rank-one formulas then agrees with the strong limit, uniquely fixing the continuum operator and its factors.

Direct rank-one trace-norm bound and mass variation

The nuclear norms of the rank-one kernels are exactly

$$\Phi(t) := \|P(t)\|_1 = \frac{\sqrt{1-e^{-2t}}}{2t}, \qquad \|P_L(t)\|_1 = \frac{\sqrt{1-e^{-2t}}}{L(1-e^{-2t/L})}.$$

Assume $L \geq \max\{1, 2\mu\}$, so $0 < a \leq \mu \leq L/2$. On $0 < t \leq L$, the floor inequality yields

$$\|e^{-\gamma t/L}P_L(t) - P(t)\|_1 \leq 10\frac{t}{L}\Phi(t), \qquad \gamma \in \{\tfrac{1}{2}, \tfrac{3}{2}\}.$$

To control the square-root singularity at $t=a$, put $g(x) = \log(\sinh x/x)$. Since $0 \leq g'(x) \leq x/2$ for $x > 0$,

$$0 \leq \delta := \log(R_L/R_a) \leq \frac{at}{8L^2} \leq \frac{1}{16},$$

and hence

$$|R_L - R_a| + |R_L^{-1} - R_a^{-1}| \leq \frac{at}{4L^2}(R_a + R_a^{-1}).$$

The essential integrable combined densities are **exactly**

$$|d_a^+| + |d_a^-| = \begin{cases} 2, & 0 < t < a, \\ 2a/\sqrt{t^2-a^2}, & t > a. \end{cases} \tag{B4}$$

By splitting $e^{-\gamma t/L}d_L P_L - d_a P$ into its step/phase and density differences, the trace-norm bounds for the **sum of both channels** are:

Contribution	Upper bound before the two-boundary $1/\pi$ factor
Step vector and phase, $0 < t \leq L$	$10a[1+\operatorname{arcosh}(L/a)]/L$
Hyperbolic density, $a < t \leq L$	$4a/L$ (conservative; a sharper $2a/L$ follows)

Contribution	Upper bound before the two-boundary $1/\pi$ factor
Continuum tail, $t > L$	$2a/L$
Lattice tail, $t > L$	$24a/L$

For the continuum tail, use $\int_L^\infty a\,[t\sqrt{t^2-a^2}]^{-1}dt = \arcsin(a/L) \le 2a/L$. For the lattice tail, set $\epsilon = a/L \le 1/2$, $z = t/L \ge 1$, and use $\log R_L \le (\epsilon/2)\coth(z/4) \le 5\epsilon/2$, $R_L - 1 \le 5\epsilon$, $\|P_L\|_1 \le 6/(5L)$, and the exponentially decaying phase factors. These reproduce the stated bound without numerical extrapolation.

Define $\mathcal{Z}_b := Z_b$ for the continuum cross-boundary difference at mass b. Consequently

$$\boxed{\|Z_{L,\mu} - \mathcal{Z}_a\|_1 \le \frac{a}{\pi L}\left[40 + 10\log\frac{2L}{a}\right].} \tag{B5}$$

To change the effective mass a to μ, rewrite the continuum channels as $H_b^\pm = \int_0^\infty b\,d_\pm(v)P(bv)\,dv$, where $d_\pm(v) = 1_{v>1}[(v+1)/(v-1)]^{\pm 1/2} - 1$. The derivative of $bP(bv)$ is a sum of three rank-one kernels, and elementary truncated-exponential moment bounds give

$$\left\|\frac{d}{db}[bP(bv)]\right\|_1 \le (1+\sqrt{2})\Phi(bv).$$

The combined density $w(v) = |d_+(v)| + |d_-(v)|$ is 2 below 1 and $2/\sqrt{v^2-1}$ above it. With $\Phi(t) \le 1/\sqrt{2t}$,

$$\frac{1}{\pi}\int_0^\infty w(v)\Phi(bv)dv \le \frac{C_{\text{ent}}}{2\sqrt{b}}, \quad C_{\text{ent}} = \frac{2\sqrt{2}}{\pi}\left(2 + \frac{\sqrt{\pi}\,\Gamma(1/4)}{2\Gamma(3/4)}\right).$$

Dominated differentiation on compact positive mass intervals is valid (the integrand is $O(v^{-1/2})$ at zero, $O((v-1)^{-1/2})$ at one, and $O(v^{-3/2})$ at infinity). Thus

$$\|\mathcal{Z}_b'\|_1 \le \frac{(1+\sqrt{2})C_{\text{ent}}}{2\sqrt{b}}, \qquad \|\mathcal{Z}_b - \mathcal{Z}_c\|_1 \le (1+\sqrt{2})C_{\text{ent}}|\sqrt{b} - \sqrt{c}|.$$

Since $0 \le \mu - a_L \le \mu^2/(2L)$, the calculation gives the claimed finite-size bound:

$$\boxed{\begin{aligned}&\|Z_{L,\mu} - Z_\mu\|_1 \le B_{L,\mu},\\ &\qquad B_{L,\mu} = \frac{a_L}{\pi L}\left[40 + 10\log\frac{2L}{a_L}\right] + (1+\sqrt{2})C_{\text{ent}}(\sqrt{\mu} - \sqrt{a_L}).\end{aligned}} \tag{B6}$$

For fixed $\mu > 0$, this is $O_\mu((\log L)/L)$. For each $M > 0$, a common bound for $0 \le \mu \le M$, $L \ge 2M$, follows from monotonicity of $a[40 + 10\log(2L/a)]$ on $(0, M]$:

$$\sup_{0\le\mu\le M}\|Z_{L,\mu} - Z_\mu\|_1 \le \frac{M}{\pi L}\left[40 + 10\log\frac{2L}{M}\right] + \frac{(1+\sqrt{2})C_{\text{ent}}M^{3/2}}{2L}.$$

At $\mu = 0$, both differences are precisely zero, not an unexamined singular endpoint.

direct $0 < s < 1/2$ Sobolev nuclear estimate

For $0 < s < 1/2$, characteristic multiplication by either half-line, hence by χ and $1-\chi$, is bounded on $H^s(\mathbb{R})$. One direct proof starts with the Gagliardo seminorm. The extra cutoff crossing term is bounded by $C_s \int |f(x)|^2 |x|^{-2s} dx$. Under Fourier transformation its bilinear positive kernel is proportional to

$$K_s(u,v) = \frac{|u-v|^{2s-1}}{|u|^s |v|^s}.$$

The weighted Schur test with weight $|u|^{-1/2}$ reduces precisely to $\int_{\mathbb{R}} |1-t|^{2s-1} |t|^{-s-1/2} dt < \infty$. The exponent conditions at $0, 1, \infty$ give $0 < s < 1/2$. There is no assertion at $s = 1/2$, where nonzero boundary traces already obstruct sharp cutoffs.

For the continuum rank-one vectors, the Fourier transforms of the zero extensions are

$$\hat{u}_t(u) \propto \frac{1 - e^{-t-iu}}{t+iu}, \qquad \hat{v}_t(u) \propto \frac{1}{t+iu}.$$

Direct integration after splitting $t \le 1$ and $t \ge 1$ gives

$$\begin{aligned} &\|v_t\|_{H^s} \le C_s (t^{-1/2} + t^{s-1/2}), \\ &\|u_t\|_{H^s} \le C_s \quad (t \le 1), \qquad \|u_t\|_{H^s} \le C_s t^{s-1/2} \quad (t \ge 1). \end{aligned}$$

Together with the elementary L^2 norms, both rank-one nuclear output norms $\|u_t\|_{H^s} \|v_t\|_2$ and $\|v_t\|_{H^s} \|u_t\|_2$ are bounded by $C_s t^{-1/2}$ **at all positive** t. The inequality $s - 1 \le -1/2$ for $t \ge 1$ is the relevant global step. Equation (B4) gives the exact scalar integral

$$\int_0^\infty (|d_\mu^+| + |d_\mu^-|) t^{-1/2} dt = \sqrt{\mu} \left[4 + \frac{\sqrt{\pi}\,\Gamma(1/4)}{\Gamma(3/4)} \right].$$

Since the left and right exterior channels, their adjoints, and the fixed finite spin matrices are all included, the rank-one Bochner integrals converge in the operator nuclear ideal $\mathcal{S}_1(L^2, H^s)$ and prove

$$\boxed{\|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,H^s)} \le C_s \sqrt{\mu}, \qquad 0 < s < 1/2, \quad \mu \ge 0.} \tag{B7}$$

This statement is about a nuclear operator **from** L^2 **into** H^s, not the eigenvalues of a single unbounded Sobolev multiplier; no commutation is required.

Finite cross-block trace path

Lemma (Finite cross-block trace path). Let P be an orthogonal projection, let χ have finite rank, and put $C = \chi P \chi$ and $X = \chi P (1-\chi)$. With

$$\mathbb{H}(X) = \begin{pmatrix} 0 & X \\ X^* & 0 \end{pmatrix}, \qquad F(t) = h\left(\frac{1+\sqrt{1-4t^2}}{2} \right), \quad h(p) = -p \log p - (1-p)\log(1-p),$$

one has $XX^* = C - C^2$, $\|\mathbb{H}(X)\| \le 1/2$, and $S(C) = \frac{1}{2} \operatorname{Tr} F(\mathbb{H}(X))$. For the embedded finite cross blocks, define

$$\mathbb{D}_{L,\mu} = \mathbb{H}(Z_{L,\mu}), \quad A_{L,\theta} = \mathbb{H}(X_{L,0}) + \theta \mathbb{D}_{L,\mu}, \quad Y_{L,\theta} = A_{L,\theta} + I/2,$$

and, in the continuum,

$$A_\theta = \mathbb{H}(X_0) + \theta \mathbb{D}_\mu, \qquad Y_\theta = A_\theta + I/2.$$

The interpolants $Y_{L,\theta}, Y_\theta$ are positive contractions and

$$\Delta S_L(\mu) = \frac{1}{2}\int_0^1 \mathrm{Tr}[F'(A_{L,\theta})\mathbb{D}_{L,\mu}]\,d\theta.$$

The analogous continuum integral $\mathcal{S}(\mu) = \frac{1}{2}\int_0^1 \mathrm{Tr}[F'(A_\theta)\mathbb{D}_\mu]d\theta$ is finite whenever $\mathbb{D}_\mu$ is trace class.

Proof. The (χ,χ) block of $P^2 = P$ gives $XX^* = C - C^2 \le I/4$. A singular-value decomposition of X gives signed eigenvalue pairs $\pm s_j$ of $\mathbb{H}(X)$. Since $F(\sqrt{p(1-p)}) = h(p)$ and $F(0) = 0$, the entropy identity follows. For the embedded finite operators the interval is infinite dimensional as an L^2 subspace, but $X_{L,\mu}$ has finite rank and the compression is zero off its finite step-function subspace, so the same argument applies. The norm bound $1/2$ is preserved by convex interpolation, proving the contraction claims. The ranges of the two finite endpoint block operators span a common finite-dimensional invariant subspace. On it F is C^1 on $[-1/2, 1/2]$, with $F'(0) = 0$, and the trace derivative is $\mathrm{Tr}[F'(A_{L,\theta})\mathbb{D}_{L,\mu}]$. This identity follows, for example, by uniformly approximating F' by polynomials and applying cyclicity to their primitives before passing to the limit in finite dimension. Integration proves the path formula. Finally $\|F'\|_\infty \le 2$, so the continuum product is trace class and its trace is continuous in θ by continuous functional calculus. The continuum integral is absolutely finite; no difference of divergent continuum entropies is taken. □

Uniform projector comparison and entropy consequence

Let $p_{\mu/L}(k)$ be the lattice Bloch projection, and $p_\mu(u)$ the continuum multiplier. On $|u| \le \pi L$, write

$$q_L(u) = \mathrm{sinc}^2(u/(2L)), \quad a = (L\sin(u/L), \mu + L[1-\cos(u/L)]), \quad b = (u, \mu).$$

The Pauli projector formula, $\|a/|a| - b/|b|\| \le 2|a-b|/|b|$, and elementary Taylor-remainder inequalities give $\|p_{\mu/L}(u/L) - p_\mu(u)\| \le |u|/(\sqrt{2}L)$, uniformly in $\mu \ge 0$. **The output can contain aliases.** Rather than dropping them, use that both full-line operators are orthogonal projections to derive the exact band-limited squared-norm identity

$$\|[\widetilde{P}_{L,\mu} - P^c_\mu]f\|_2^2 = \int q_L \|[p_{\mu/L}(u/L) - p_\mu(u)]\hat{f}(u)\|^2 du + \int (1-q_L)\|p_\mu(u)\hat{f}(u)\|^2 du.$$

Since $1 - q_L \le u^2/(12L^2)$ on the band and the difference of orthogonal projections has norm at most one off it, frequency splitting proves

$$\boxed{\|\widetilde{P}_{L,\mu} - P^c_\mu\|_{H^s \to L^2} \le C_s L^{-s}} \qquad (0 < s < 1/2), \tag{B8}$$

with C_s independent of L, μ. The cutoff multiplier lemma supplies the same estimate for the self-adjoint off-diagonal interpolants, and their continuum versions act boundedly on H^s.

For the entropy branch, set $F(t) = h((1+\sqrt{1-4t^2})/2)$, $h(p) = -p\log p - (1-p)\log(1-p)$, and $g(z) = F'(z/2)$. The exact identity

$$g(z) = -2z\log|z| + r(z), \quad r \in C^2([-1,1]),$$

implies that the cosine–Chebyshev coefficients of g are $O(k^{-2})$. For the logarithmic term the odd coefficients at $k = 2j+1 \ge 3$ are exactly $(-1)^j/[j(j+1)]$, with even coefficients zero. Thus the degree-n Chebyshev truncation approximates g uniformly within C/n; when written as a polynomial in $Y \in [0,1]$, its monomial coefficient ℓ^1-norm is at most $C7^n$.

For each θ, abbreviate $Y_L = Y_{L,\theta}$ and $Y = Y_\theta$ from the trace-path lemma. The noncommutative telescoping identity

$$(Y_L^k - Y^k)\mathbb{D}_\mu = \sum_{j=0}^{k-1} Y_L^j (Y_L - Y) Y^{k-1-j} \mathbb{D}_\mu$$

places the **only** high-regularity input to the right of the $H^s \to L^2$ difference. Left lattice factors are merely L^2 contractions, so no unjustified lattice H^s functional calculus is used. Taking the nuclear norm and applying (B7)–(B8) yields

$$|\Delta S_L(\mu) - \mathcal{S}(\mu)| \le 2B_{L,\mu} + \frac{C}{n}\|\mathbb{D}_\mu\|_1 + C_s n (7M_s)^n L^{-s} \|\mathbb{D}_\mu\|_{\mathcal{S}_1(L^2,H^s)}.$$

Choosing $s = 1/4$ and $n = \lfloor s \log L / [2\log(7M_s)] \rfloor$ (for sufficiently large L) establishes

$$\boxed{\sup_{0\le\mu\le M} |\Delta S_L(\mu) - \mathcal{S}(\mu)| = O_M(1/\log L)}. \tag{B9}$$

Here $\mathcal{S}(\mu)$ is the finite continuum **trace path**, not the difference of two separately divergent continuum entropies. The finite trace-path lemma above proves the spectral-calculus and singular-value identities used in this assertion. Thus (B9) is a theorem for the specified boundary subsystem; it does not assume a full relative-entropy finite-part limit.

Scope and limitations

The estimates concern restrictions of infinite-chain quasifree projections, not the occupied projector of an open finite chain. The derived boundary estimates and entropy-difference rate do not by themselves imply modular or nonlinear convergence. Those separate inputs are proved for this model in *Finite Part of Wilson–Dirac Relative Entropy*, whose Theorem 1 assembles the full fixed-μ expansion, and are sharpened in *Uniform Modular Control and Nonlinear Convergence for Wilson–Dirac Intervals*. No uniform Coulomb or moving-edge matching estimate is asserted as a consequence of the boundary argument.

Mathematical references

1. NIST Digital Library of Mathematical Functions, §§5.12 and 10.32 (beta integrals and modified-Bessel integral representations), https://dlmf.nist.gov/5.12 and https://dlmf.nist.gov/10.32.
2. R. Frank and R. Seiringer, "Non-linear ground state representations and sharp Hardy inequalities," *Journal of Functional Analysis* 255 (2008), 3407–3430, doi:10.1016/j.jfa.2008.05.015 (fractional Hardy background).

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